



AI Hackathon 2026

Port Operations Decision-Support Solution

Container Congestion Predictor

**AI-Powered Proactive Port
Operations Optimisation**

Presented By:

AI FALCONS

Predict Early. Optimise Continuously. Act Fast.

The Critical Problem

Port congestion is costing time, money, and operational efficiency

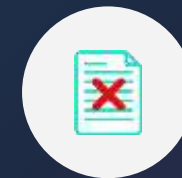
Thynk Unlimited



Port congestion causes vessel queues, delayed berth assignments, and cascading supply chain disruptions.



Hotspot detection is reactive — congestion is only identified after delays have already escalated.

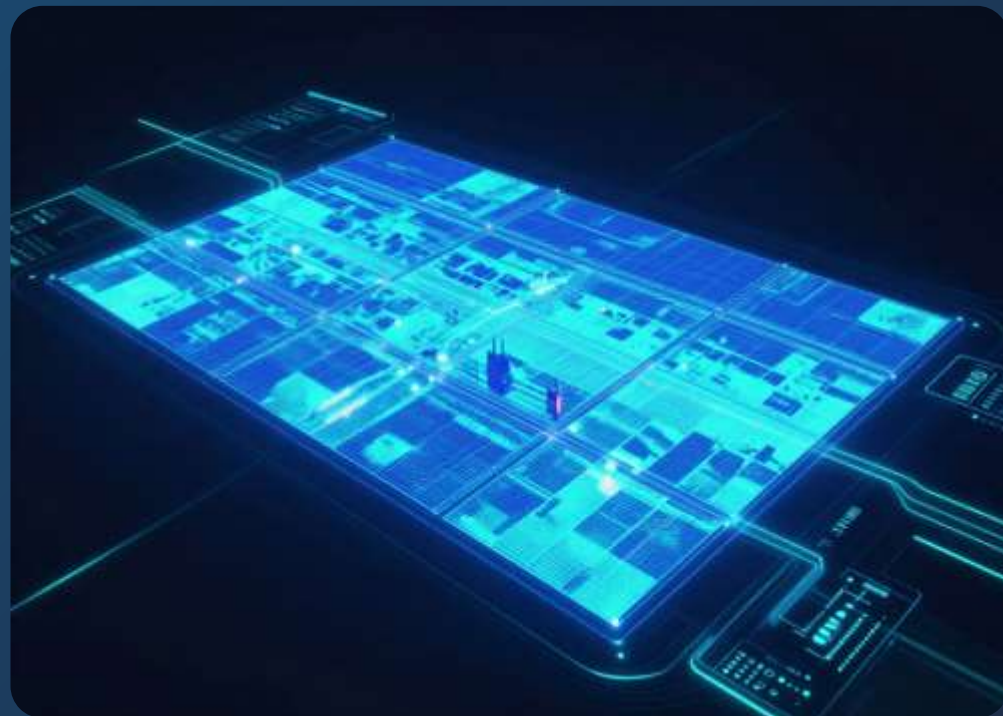


Manual spreadsheets and siloed data make real-time coordination across berths and cranes nearly impossible.



Late routing decisions increase waiting times, raise fuel costs, and reduce crane utilisation.

Our Solution



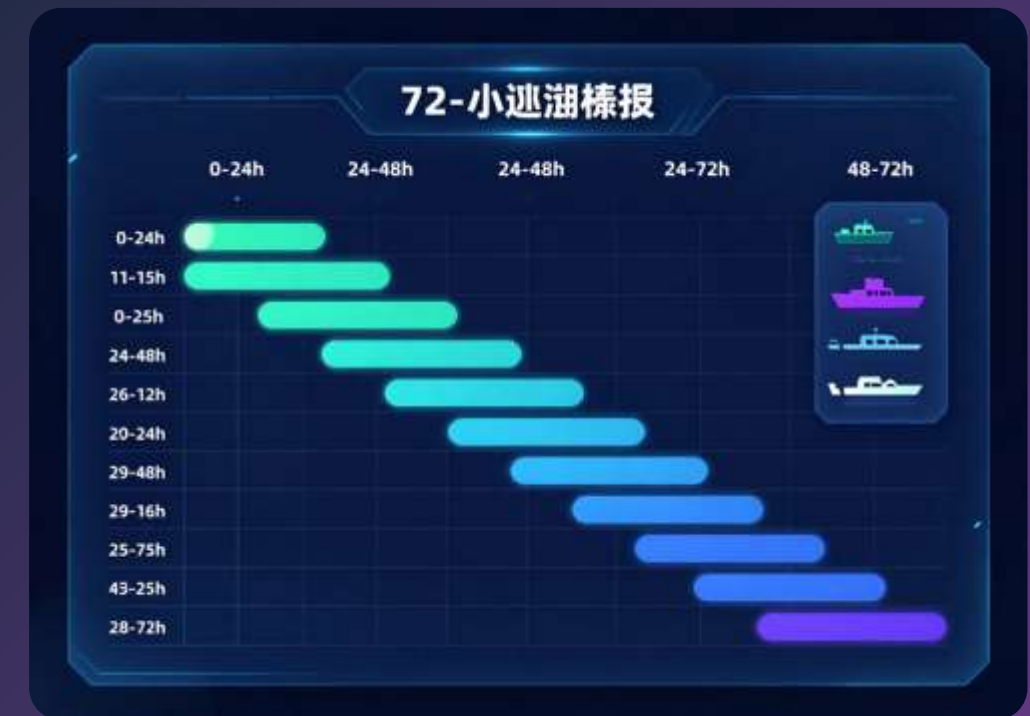
Predict & Detect

AI forecasts berth congestion 6 hours ahead and detects emerging hotspots in real time.



Optimise & Reroute

Engine minimises vessel wait times and dynamically reroutes ships to alternate berths or terminals.

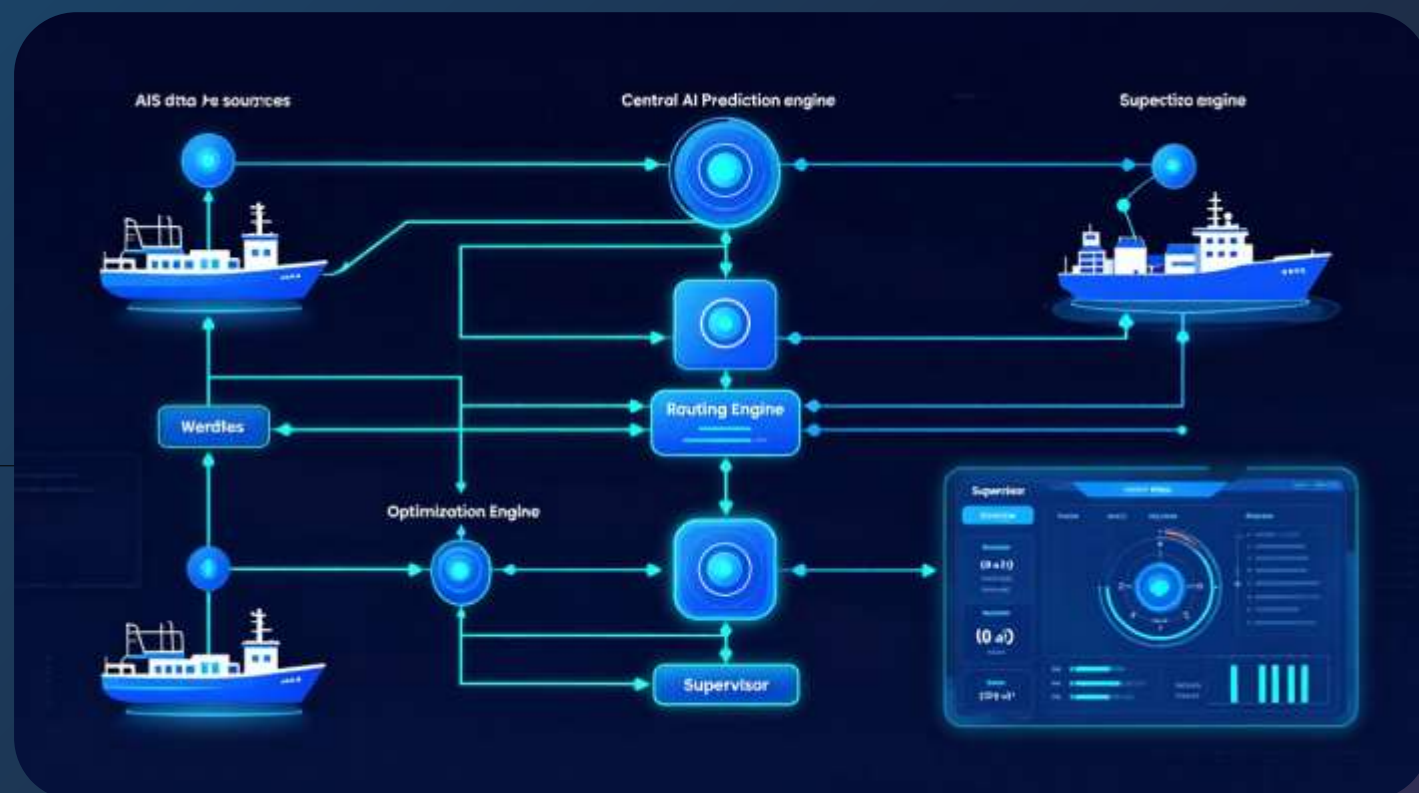


Plan

Generates a rolling 72-hour operations plan with prioritised actions for supervisors.

System Architecture

Integrated data pipelines powering real-time port intelligence



Data Ingestion Layer

Vessel schedules, AIS streams, berth and crane sensor data feeds.

Prediction Engine

6-hour slot congestion scoring, berth pressure and arrival clustering.

Optimisation & Routing

Berth allocation, crane scheduling and alternate terminal routing engine.

Supervisor Dashboard

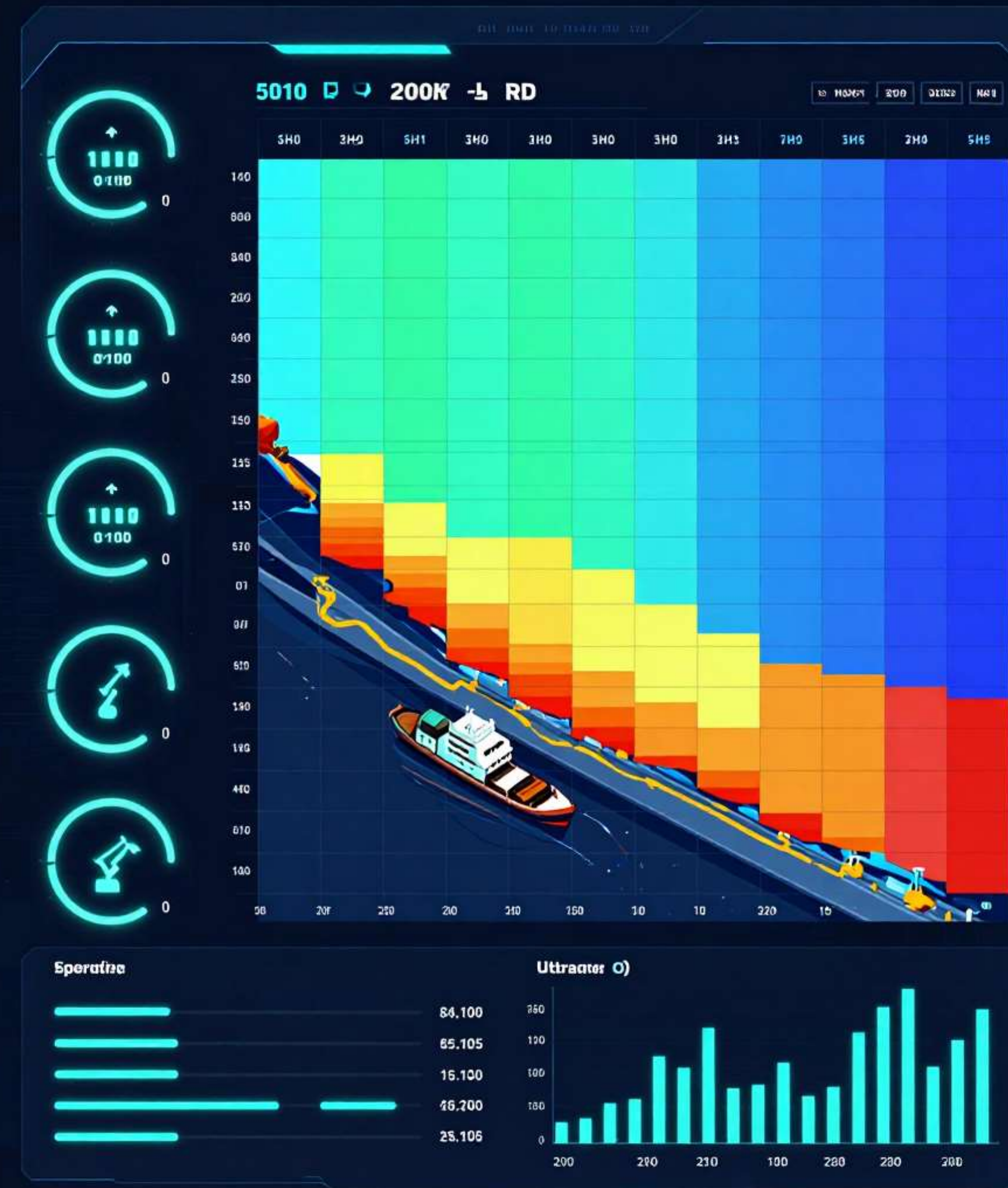
Real-time alerts, vessel assignments, KPIs and 72-hour operations plan.

Container Congestion Predictor

Congestion Prediction

6-hour time slots forecast berth pressure, crane utilisation, and vessel arrival clustering. Each slot generates a 0–100 risk score, enabling proactive intervention before congestion peaks.

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Berth & Crane Optimisation

Minimise Waiting. Maximise Throughput.

Allocates berths and cranes to minimise vessel waiting time subject to capacity, priority, and operational constraints — delivering optimal, conflict-free scheduling across all assets.



Container Congestion Predictor

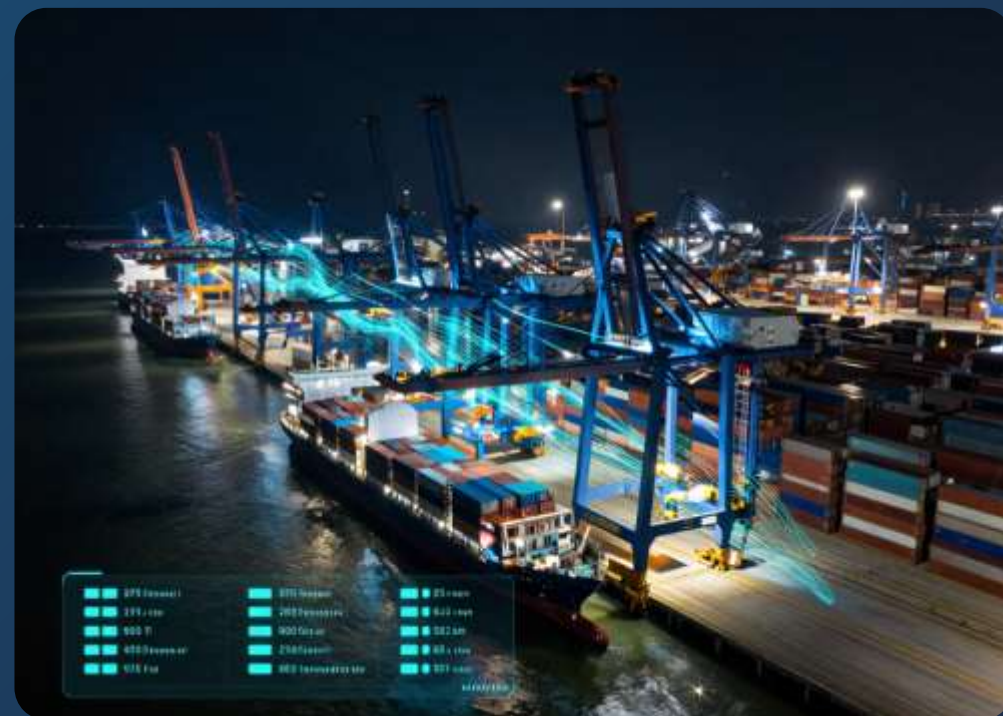
Alternate Routing

Compare preferred vs. alternate berths and terminals across three key penalty dimensions:

- ⚓ Wait Penalty
- Queue delay at preferred berth
- Congestion Penalty
- Live risk score 0–100
- ↔ Rerouting Penalty
- Cost of switching terminal



72-Hour Operations Plan



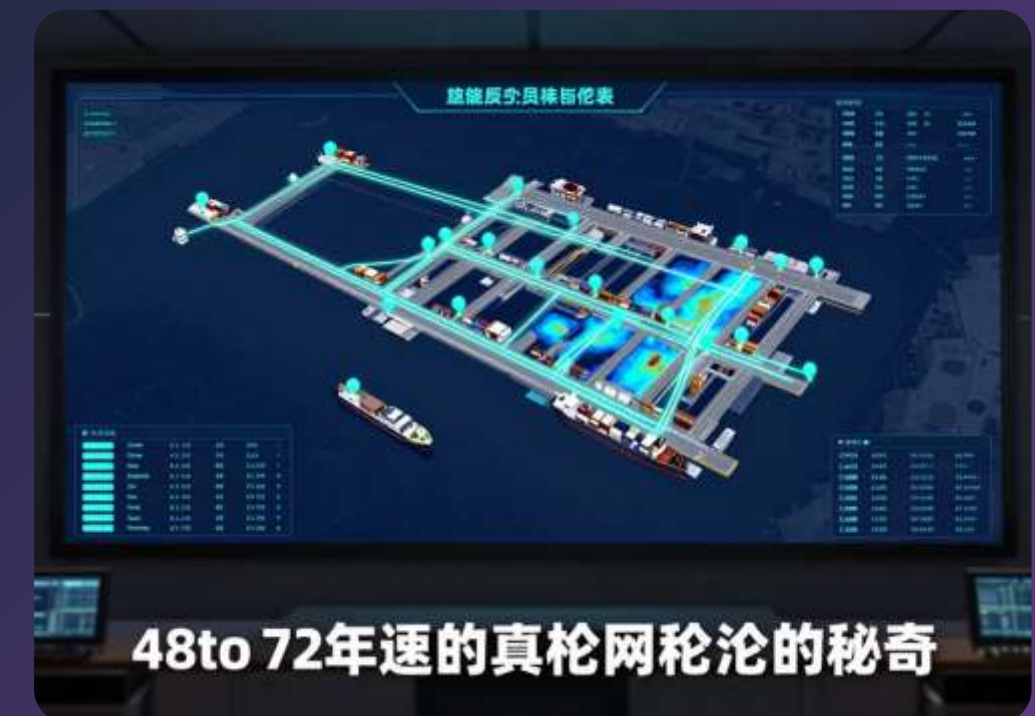
0–24 Hours

Immediate alerts, vessel rerouting decisions, berth reassignments, and crane reallocation for arriving ships.



24–48 Hours

Proactive berth scheduling, optimised crane deployment, and congestion mitigation for incoming vessel queue.



48–72 Hours

Strategic capacity planning, pre-positioning resources,
and coordinating multi-terminal routing for peak
readiness.

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Technology Stack

Core components powering the port operations decision-support system



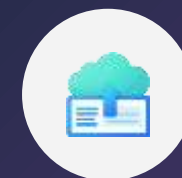
Python Prototype: Modular pipeline for ingestion, prediction, optimisation, and routing logic.



AIS / TOS Data: Live vessel tracking and terminal operating system feeds as core inputs.



Explainable Scoring: Transparent risk scores with factor breakdowns for operator trust.



Dashboard & Cloud: Interactive supervisor UI with cloud deployment for real-time access.

Expected Impact

Measurable outcomes from AI-powered port operations optimisation



Up to 40% earlier congestion warnings, enabling proactive decisions before queues form.



20% improvement in crane utilisation through optimised scheduling and allocation.



30–50% reduction in average vessel waiting time at berths and anchorage.



Faster operational decisions with human-in-the-loop supervisor dashboard alerts.

Demo Flow & Future Scope



Live Demo Flow

Load data → Predict congestion → Detect hotspots →
Optimise berths → Reroute vessels → Generate 72-hour
plan.



Future ML & Data

Advanced ML models, live weather integration, yard and
gate data feeds, predictive crane scheduling.



Scale & Expansion

Multi-port routing network, cloud deployment, truck and rail
coordination, global port interoperability.



Container Congestion Predictor



Predict early.
Optimise
continuously.
Act before congestion.

AI-powered proactive port operations — transforming reactive chaos into intelligent, data-driven decisions for the ports of tomorrow.



Team: AI FALCONS 2026



Port Ops AI Solution

Presents Solution

THANK YOU

CREATED & PRESENTED BY:-
TEAM AI FALCONS